

# SANKUR KUNDU

## Software Engineer | Backend, Systems & Full-Stack

Bhubaneswar, India | +91 97761 94201 | [sankur.kundu.tw@gmail.com](mailto:sankur.kundu.tw@gmail.com)  
[linkedin.com/in/sankur-kundu](https://linkedin.com/in/sankur-kundu) | [github.com/Shakhtar-Sankur](https://github.com/Shakhtar-Sankur) | [shakhtar-sankur.github.io](https://shakhtar-sankur.github.io)

## SUMMARY

Software engineer (B.Tech CSE, 2026) who builds systems from scratch and proves them correct: **relay** (an LLM inference cluster in **C++/CUDA** and **Swift**), **tandem** (distributed training **bit-identical to PyTorch**), **quorumbd** (TLA+-verified SQL), **tessel** (GPU kernels at **cuBLAS parity**). Co-founded Gigzen as its sole engineer.

## PROJECTS

### relay — C++, CUDA and Swift: a disaggregated LLM inference cluster

- Built **prefill** and **decode workers** in **C++/CUDA** (paged KV cache, continuous batching, **prefix caching**), a **KV-cache transfer engine** that streams each layer while the next computes, and a **Swift** control plane (**Swift-C++ interop**, OpenAI-compatible streaming API, **prefix-aware routing**, admission control).
- On a **T4**, layer streaming hides **99–100%** of the KV transfer (46 MB: **21.6 ms** → **0.08 ms**); outputs match **Hugging Face** and stay **bit-identical** whether batched, cached, preempted or split across workers; **ThreadSanitizer**-clean.

### tandem — Python, C++ and CUDA, no torch.distributed or NCCL: distributed training from scratch

- Built the parallelisms behind large-model training on its own **ring all-reduce**: **DDP** with gradient bucketing, **ZeRO-1/2/3** (**FSDP**-style sharding), **GPipe** and **1F1B** pipelines; a **C++/CUDA engine** (lock-free shared-memory rings, CUDA IPC, a fused reduce kernel) runs all-reduce **1.6–2.8× faster** than its Python engine on a **T4**.
- On **2× T4**, **DDP** and **ZeRO-1/2/3** train **bit-identically** to **PyTorch DDP/FSDP** on **NCCL**; all-reduce **within 2% of NCCL's bandwidth** at 64–256 MB; on an 86M-parameter GPT, **ZeRO-3** cuts per-GPU memory **17%**, **1F1B 32–52%** vs **GPipe**.

### quorumbd — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible

- Built from scratch a **distributed SQL database** that **psql** connects to: **LSM storage**, **Raft**, a self-rebalancing **multi-Raft** store, **serializable ACID transactions** (Percolator), a SQL planner; **1,500 tpmC** on a **TPC-C**-derived benchmark.
- Proved by **deterministic simulation** (**10,000 Raft clusters**, **610K crashes**, histories machine-checked) and a **TLA+** model (**337K states**); found **6 real bugs**, catches all **15 planted**; **time-travel debugger** rewinds runs by replay.

### Also built: tessel • faultline • kiln • ferrolm

- tessel**, a tile language for GPUs and TPUs: **GEMM at parity with cuBLAS** on a **T4**; **faultline**, Jepsen-style fault injection that tested **4 etcd release lines** (**1.26M** operations under **288** faults); **kiln**, an ML compiler **faster than torch.compile** on CPU; **ferrolm**, an LLM server.

## OPEN SOURCE

PR submitted to **PyTorch** (under review): [pytorch/pytorch#199441](https://github.com/pytorch/pytorch/pull/199441) removes **13 stale torch.compile (Inductor) test skips** (most cited a bug fixed in 2024) after checking each test now passes on **CPU** and **2× T4 GPUs**.

## EXPERIENCE

### Gigzen — Co-founder & Founding Engineer

Apr 2026 – Present

Bhubaneswar, India • Early-stage startup (incorporation in progress) • Sole engineer across every product

- Sole engineer** on **Waggle**, a delivery platform of **three Android apps** (riders, shops, customers) from one **TypeScript / React / Capacitor** codebase on **Supabase PostgreSQL**: **53 tables**, **90 RLS policies**, **154 SQL functions**.
- Put the business rules in the database, where no client can bypass them: **first-accept-wins dispatch** (45-second offer window), **4-digit handover codes**, **Razorpay/UPI** payments reconciled by **UTR**, **GST invoices** and **pg\_cron** billing runs.
- Security-tested** it with **150 automated attacks** on its database access rules (all blocked) and a **42-step Playwright** run.
- Built **Populace**, an open-source multi-user testing tool (**Node.js**, an **npm** CLI, **GitHub Action** and **Electron** app); its **first run** found **5 defects in 3.5 minutes** after manual testing had passed, one breaking **every** **signup**.

## SKILLS

**CS Fundamentals**: Data Structures & Algorithms, System Design, Distributed Systems, DBMS internals, OS, Networks

**Languages**: C++, Rust, Python, Swift (C++ interop), TypeScript, JavaScript, SQL (PL/pgSQL), Java

**Backend & Data**: PostgreSQL, Supabase, Row-Level Security, pg\_cron, REST APIs, Node.js, FastAPI, Flask, Redis, Razorpay/UPI

**Frontend & Mobile**: React, Capacitor (Android), Electron, Leaflet/MapLibre maps, HTML, CSS

**Systems & performance**: CUDA (tensor cores), LLM serving (disaggregated prefill/decode, KV-cache transfer), DDP/ZeRO, SIMD

**Testing, DevOps & Cloud**: Simulation testing, fault injection, TLA+, Playwright, GitHub Actions, Docker, Kubernetes, GCP

## EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata

Sep 2022 – Apr 2026